Set builder notation

1 List all the elements of the following sets:

a $\{\text{positive odd numbers less than } 10\}$

b $\{\text{positive multiples of } 4 \text{ that are less than } 46\}$

c $\{\text{positive even integers that are less than } 16\}$

d $\{\text{positive factors of } 42\}$

e $\{\text{square numbers less than } 109\}$

f $\{\text{numbers that can be rolled on a standard die}\}$

g $\{x | x \text{ is a natural number less than } 5\}$

h $\{x | x \text{ is an odd whole number less than } 13\}$

i $\{x | x \text{ is an integer between } -8 \text{ and } -3 \text{ (not inclusive)}\}$

2 Determine whether the following statements are true or false:

a $3 \in \{x | x \text{ is a rational number}\}$

b $3 \in \{2, 4, 5, 8\}$

c $1 \in \{9, 8, 6, 5, 1\}$

d $-0.25 \in \{x | x \text{ is a natural number}\}$

3 Consider the set $A = \{2, 4, 6, 8\}$. Construct a set builder notation for A.

Special number sets

4 Determine whether the following statements are true or false:

a $-6 \notin \mathbb{Z}$

b $\dfrac{5}{2} \in \mathbb{R}$

c $\pi \in \mathbb{N}$

d $\sqrt{2} \in \mathbb{Z}^+$

5 Determine whether the following statements are true or false:

a $\mathbb{N} \subset \mathbb{R}$

b $\mathbb{Z}^+ \subseteq \mathbb{Z}$

c $0, 1, 5, -9 \not\subseteq \mathbb{R}$

d $\mathbf{Q} \subseteq \mathbb{R}$

e $\mathbb{R} \subseteq \mathbb{N}$

f $\mathbb{Z} \subseteq \mathbb{N}$

g $\mathbb{Z}^+ \subset \mathbf{Q}$

h $\mathbb{N} \subseteq \mathbb{N}$

6 Complete the following table to indicate whether each number belongs in the number set:

	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Z}^+$	$\mathbf{Q}$	$\mathbb{R}$
5	Y	Y	Y	Y	Y
-3	N	Y	N	N	Y
123					
-94.5					
$\frac{3}{4}$					
$\sqrt{5}$					
$\sqrt{-1}$					